

培道中學校

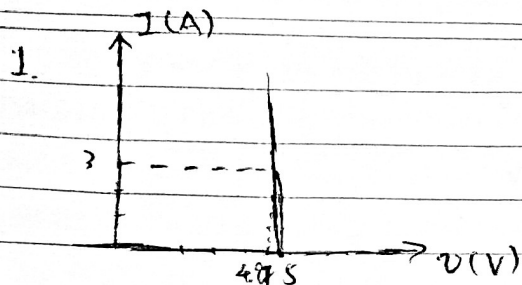
姓名：

(號)

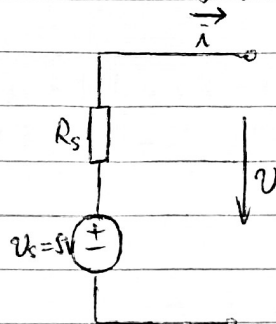
年級：

科目：

成績

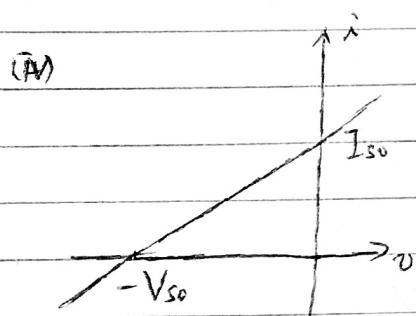
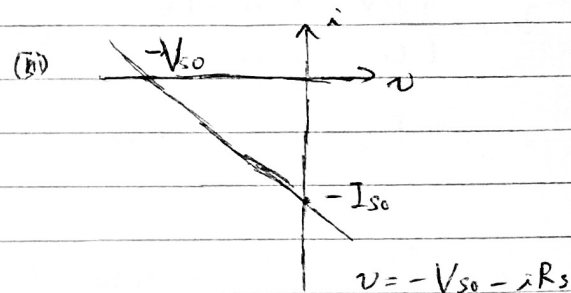
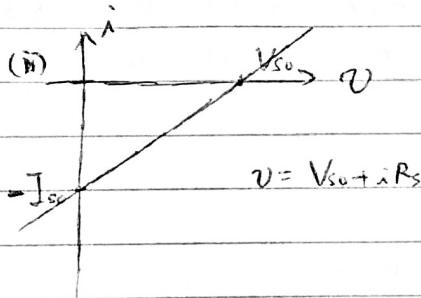
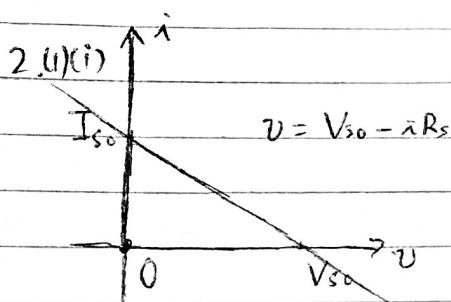


因為由圖中可見，在電源正常工作範圍內
 $v-i$ 曲線可近似為直線。

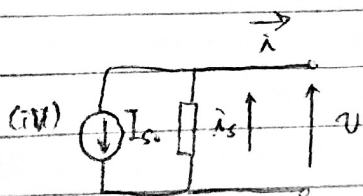
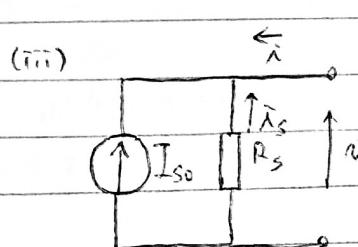
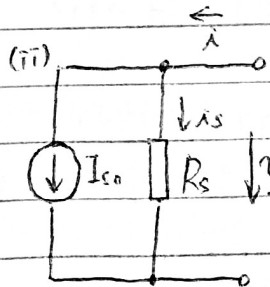
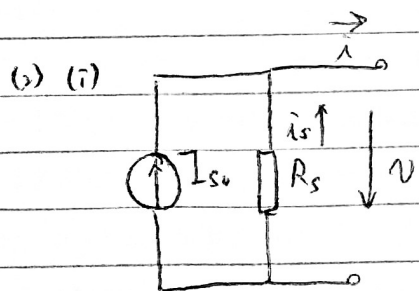


$$v = v_s - i R_s$$

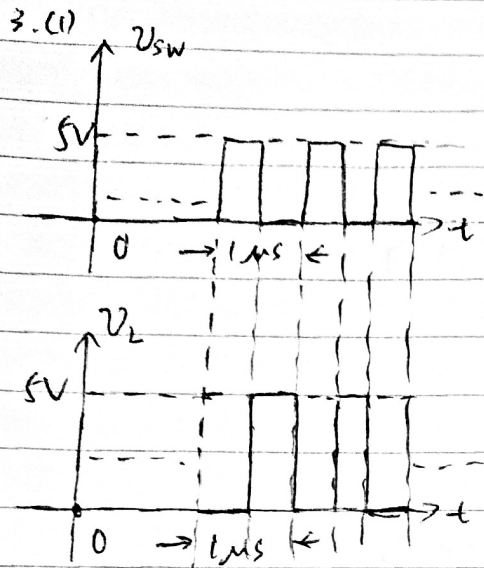
$$R_s = \frac{5 - 4.97}{3} = 0.03 \Omega$$



(i), (iii) 一、三象限釋放能量
 (ii), (iv) 二、四象限釋放能量



伏安特性曲線與 (i) 中四圖對應一致
 端口伏安特性不與參考電壓、電流方向相關



(2) $\overline{v_{DC}} = \sqrt{\overline{v_L^2}} = v_{DC} = \overline{v_L(t)} = 2.5V$

(3)
$$P(t) = \begin{cases} 0W, & v_c(t) = -1V \\ 0.5W, & v_c(t) = 1V \end{cases}$$

(4) $\overline{P} = \overline{P(t)} = 0.5 \times \frac{1}{2} = 0.25W$
 $v_{rms} = \sqrt{\overline{P} R} = \frac{5}{\sqrt{2}} = 3.536V$

(5) $P_{sw} = v_c i_c + v_{sw} i_{sw} = 0 + v_{sw} i_{sw} = v_{sw} i_L$

$$v_{sw} = \begin{cases} 0V, & v_c(t) = 1V \\ 5V, & v_c(t) = -1V \end{cases}, \quad i_L = \begin{cases} 0.1A, & v_c(t) = 1V \\ 0, & v_c(t) = -1V \end{cases}$$

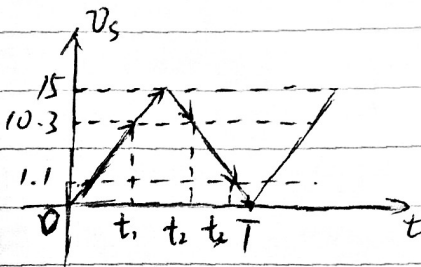
 $\Rightarrow P_{sw} = v_{sw} i_L = 0W$

(6) $P_{DC} = \frac{v_{DC}^2}{R} = \frac{6.25}{50} = 0.125W, \quad P_{AC} = \overline{P} - P_{DC} = 0.125W$

4.

$$v_D = \begin{cases} \frac{10}{3} k\Omega \cdot i_D, & 0 \leq i_D \leq 3mA \\ -9.3 k\Omega \cdot i_D + 37.9V, & 3mA < i_D \leq 4mA \\ 0.7V, & i_D > 4mA \end{cases}$$

$$v_s = v_D + v_L = \begin{cases} \frac{103}{30} k\Omega \cdot i_D, & 0 \leq i_D \leq 3mA \quad \in [0, 10.3]V \\ -9.2 k\Omega \cdot i_D + 37.9V, & 3mA < i_D \leq 4mA \quad \in [1.1, 10.3]V \\ 0.7V + 100\Omega \cdot i_D, & i_D > 4mA \quad \in (1.1, +\infty)V \end{cases}$$



因為其週期性 (v_s 的兩週期),
 所以我們僅需考慮一個週期的情況便可。

如左圖, 標記了每個時間點 t_1, t_2, t_3

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(1) 當 $0 \leq t \leq t_1$ 時, 有 $0 \leq i_D \leq 3 \text{ mA}$

$$\Rightarrow v_L = i_L R_L = i_D R_L = \frac{v_s}{\frac{103}{30} \text{ k}\Omega} \cdot 100 \Omega = \frac{3}{103} v_s(t), \quad v_D = \frac{102}{103} v_s$$

(2) 當 $t_1 < t \leq t_2$ 時, 有 $v_s \in (10.3, 15]$, i_D

$$\Rightarrow v_s = 0.7 \text{ V} + 100 \Omega \cdot i_D, \quad i_D \in [96, 143] \text{ mA}$$

$$\Rightarrow v_L = i_D R_L = \frac{v_s - 0.7 \text{ V}}{100 \Omega} \cdot 100 \Omega = v_s - 0.7 \text{ V}, \quad v_D = 0.7 \text{ V}$$

(3) 當 $t_2 < t \leq t_3$ 時, 有 $4 \text{ mA} \leq i_D \leq 96 \text{ mA}$

$$\Rightarrow v_s = 0.7 \text{ V} + 100 \Omega \cdot i_D \Rightarrow v_L = v_s - 0.7 \text{ V}, \quad v_D = 0.7 \text{ V}$$

(4) 當 $t_3 < t \leq T$ 時, 有 $v_s \in [0, 1.1)$

$$\Rightarrow v_s = \frac{103}{30} \text{ k}\Omega \cdot i_D \Rightarrow v_L = \frac{3}{103} v_s, \quad v_D = \frac{102}{103} v_s$$

